# Cheat sheets

Each card has two cheat sheets, as in the online game: the experts' data on the consequences, and on how often and how much we meet it. Look only when a card seems too hard. If you keep score, each cheat sheet used takes off a quarter of the card's points.

↑ HOW SERIOUS → HOW OFTEN, HOW MUCH

## 01 Campylobacter

↑ Most recover in a couple of weeks; rarely, Guillain-Barré syndrome follows

→ They reach us now and then: through an undercooked piece, splashes or a dirty board

## 02 Salmonella

↑ Passes in a few days; more dangerous for small children and older people

→ Eggs and meat are eaten almost daily, but control programmes and heat keep Salmonella off the plate

## 03 Listeria monocytogenes

↑ The highest share of patients who die of any food infection in the European Union

→ Listeriosis is about 55 times rarer than Campylobacter infection: ready meals are eaten by the date, before much Listeria builds up

## 04 Norovirus

↑ Fatal cases are very rare and mainly affect already frail patients

→ Highly infectious; outbreaks are regular, and not only through food

## 05 Toxoplasma gondii

↑ Harmless in a healthy adult; effects on a foetus can be permanent

→ Heat or freezing kills the parasite, but it survives in cured meat and on unwashed vegetables

## 06 Shiga toxin-producing E. coli

↑ In some children it causes kidney failure with lasting damage

→ A handful of bacteria is enough, but an ordinary kitchen cooks mince through, and heat kills them

## 07 Yersinia enterocolitica

↑ Usually clears up on its own, but sometimes leads to needless surgery

→ In an ordinary kitchen pork gets cooked, and heat kills Yersinia

## 08 Aflatoxins

↑ There is no safe dose: in theory any amount raises the risk

→ The limit is as low as technically possible, and contaminated shipments are stopped at the European Union border

## 09 Deoxynivalenol (DON)

↑ Effects appear at high doses and pass within hours

→ Not an exotic toxin: it comes with everyday bread and porridge, and children's average intake exceeds the daily limit

## 10 T-2 and HT-2 toxins

↑ Immune effects are seen even at comparatively low doses

→ Mainly people who eat oat flakes take them in, and since 2024 oats have binding limits in the European Union

## 11 Ochratoxin A

↑ Harms the kidneys with long-term intake; carcinogenic in animals

→ Everyday products usually hold little ochratoxin, and only years of intake harm the kidneys, not a single loaf

## 12 Patulin

↑ The irritation passes, and the limit is set by it alone

→ Juice is pressed from sound fruit; only a lot of juice from bruised apples would reach the limit

## 13 Alternaria toxins

↑ In cell studies they damage DNA; assessors assume a cancer risk as a precaution, unproven in humans

→ The European Food Safety Authority (EFSA) estimated that our long-term intake exceeds the threshold below which there is no concern

## 14 Acrylamide

↑ Carcinogenic in animals, only probably in humans; no acute effects, the lifetime dose is what counts

→ Almost everyone takes it in every day, only a few hundred times less than the dose that caused tumours in animals

## 15 Furan

↑ Liver damage in animals; the margin to a harmful dose is small

→ Coffee is drunk daily, but furan is volatile: most of it escapes while the coffee is made and cools

## 16 Benzo(a)pyrene and PAHs

↑ One meal does no harm: the effect adds up over a lifetime

→ Depends on habits: smoked food carries polycyclic aromatic hydrocarbons (PAHs), but not everyone eats it, and not every day

## 17 Glycidyl esters

↑ Harm can only build up over a long time; infants are the most at risk

→ Formula-fed infants take in the most, but their food has the strictest limit

## 18 Lead

↑ No safe level, but everyday intake is low: the effect is small, though lasting

→ Little in each source, but many sources: the doses add up to a high overall exposure

↑ HOW SERIOUS → HOW OFTEN, HOW MUCH

### 19 Cadmium

↑ The safe level has a wide margin: even a little above it there is no harm yet; in smokers it builds up much faster

→ Everyone takes some in every day, and it builds up in the kidneys for decades: years count, not days

### 20 Inorganic arsenic

↑ No safe intake level can be set; what is measured is the gap to the dose at which cancer was seen in humans

→ In Europe exposure is far lower than in South Asia, where arsenic from underground gets into drinking water

### 21 Methylmercury

↑ The unborn child, whose nervous system is developing, is the most sensitive; in adults the effect is smaller

→ Mostly in predatory fish, which are eaten comparatively rarely in Latvia

### 22 PFAS ("forever chemicals")

↑ The limit is set by how well vaccines work in children; the substances can harm health over a long time

→ They do not break down and leave the body very slowly, so doses from many sources build up

### 23 Dioxins and dl-PCBs

↑ The most sensitive effect: a lower sperm count in men highly exposed as boys; they stay in the body for years

→ They build up in fat and stay in the body for years, so doses from milk, meat, eggs and fish add up

### 24 Histamine

↑ Symptoms start quickly and pass within hours

→ The cold chain stops it: fish kept cold does not form it, but cooking will not remove it

### 25 Nitrate

↑ Harmful effects are only seen at doses far above the everyday intake

→ Most in winter-grown leafy vegetables; adults regularly go over the acceptable daily intake

### 26 Pesticide residues

↑ Residues are usually far below the dose at which effects are seen

→ Fruit and vegetables are eaten every day, but the residues found are nearly always below the legal maximum

### 27 Glutamate (E621)

↑ Reactions to a large dose usually pass quickly, with no lasting effects

→ For an adult the daily limit is about 2 g, but children can exceed it even with an average diet

### 28 Peanut allergy

↑ For an allergic person a tiny amount is enough: traces count, not quantity

→ Only allergic people, and labels protect them: the danger is a mislabelled pack

### 29 Aspartame

↑ You would have to drink a great many cans to reach the daily limit

→ Only people who regularly use sugar-free products take it in, and even they are usually far from the daily limit

### 30 Microplastics

↑ Toxicology data are missing: that means neither that microplastics are safe nor that they are dangerous

→ We use water and salt every day, but no authority can yet say how much we take in

### 31 Titanium dioxide (E171)

↑ DNA damage cannot be ruled out, which is exactly why it was banned

→ It is still taken in with medicines, but without a safe dose nobody can say whether that is a lot or a little

### 32 Cyanogenic glycosides

↑ Eating too much usually brings nausea and headache; severe poisoning is rare

→ These substances are only in a few plants, mainly in seeds and kernels; whole apple seeds pass through intact

### 33 Undercooked chicken

↑ Most recover in a couple of weeks; rarely, Guillain-Barré syndrome

→ Chicken is eaten almost weekly; below 74 °C in the centre, every infected bird reaches the plate with live bacteria

### 34 Home-slaughtered game

↑ Most fall seriously ill, not just a few: most end up in hospital for weeks

→ Game is eaten a few times a year, but freezing does not kill Trichinella: a test or thorough cooking protects

### 35 One board for meat and salad

↑ The same bacteria as in raw chicken; death and lasting nerve damage are the exception

→ It happens unnoticed, and salad is never heated: bacteria from raw meat stay alive on it

### 36 Unwashed hands

↑ A few rough days; no lasting effects in a healthy person

→ Hands touch food several times a day, and about 18 virus particles are enough

### 37 Cooked too brown

↑ No acute effect, and it is only probably carcinogenic in humans: this is about a lifetime's intake

→ A dark crust adds more every time, and per kilogram of body weight children take in the most

### 38 Washing chicken before cooking

↑ The same consequences as undercooked chicken, delivered elsewhere

→ The bird itself gets cooked: bacteria reach people only through splashes that fly about half a metre

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### 39 Unwashed salad and berries

↑ Usually mild; the severe cases involve Shiga toxin-producing Escherichia coli (STEC) and small children

→ Contamination comes from the field, and salad and berries are eaten raw: only washing lowers the germ count

### 40 Frozen berries straight from the freezer

↑ Norovirus, gone in a few days, sets the rating, not hepatitis A

→ Frozen in summer, eaten in winter: the virus survives the whole way, but one minute of boiling kills it

### 41 Raw milk

↑ In small children Shiga toxin-producing Escherichia coli (STEC) can cause kidney failure with lasting damage

→ Few people drink it, but they drink it regularly, and boiling deals with everything the milk may carry

### 42 Mushrooms you picked yourself

↑ At first you seem to get better; that is when the toxins destroy the liver

→ Many pick mushrooms, but mostly in autumn, and cooking does not destroy the toxin: only sure identification protects

### 43 Undercooked burger

↑ 60% of haemolytic uraemic syndrome cases in the European Union are in children under five

→ Mince is eaten several times a month, and in a pink middle the bacteria stay alive

### 44 Energy drinks

↑ What years of use do to teenagers is not known: there are too few studies, and the European Food Safety Authority (EFSA) is reassessing it

→ One can exceeds the figure the European Food Safety Authority (EFSA) gives for a 40 kg child, but that figure was carried over from adults

### 45 Leftovers left out

↑ Usually over within a day; very rarely the toxin damages the liver

→ Spores survive boiling and the toxin survives reheating, so time decides, not heat

### 46 Thawing meat on the counter

↑ Thorough cooking kills the bacteria, but not all of their toxins

→ While the middle is still hard, the surface sits for hours in the warm, where bacteria grow fastest; in a fridge they barely do

### 47 Frozen breaded chicken

↑ Usually over in a week, but young children and older people may need hospital care

→ The crumb is deceptive, and in Canada about 40 % of shoppers think they are precooked; what decides is whether the raw meat carries Salmonella

### 48 Frozen vegetables, uncooked

↑ In a European outbreak, 9 of the 47 patients died

→ Blanching is not cooking, but at home frozen vegetables are usually boiled, and heat kills Listeria

### 49 Raw dough

↑ In children, Shiga toxin-producing Escherichia coli (STEC) can cause haemolytic uraemic syndrome; in the European Union in 2023, 60% of those cases were in children under five

→ Bacteria cannot grow in dry flour but survive, and only baking kills them; everyone wants to lick the spoon

### 50 Eating food past its “use by” date

↑ Listeriosis patients almost always need hospital care; about one in six dies

→ Food past the date is seldom eaten, and listeriosis is rare: about 55 times rarer than Campylobacter infection

### 51 Eating food past its “best before” date

↑ A quality date: if the pack is intact and it looks and smells fine, it can be eaten

→ This date is on long-keeping staples found in almost every home; past it quality changes, not safety

### 52 Scraping off the mould

↑ Mycotoxins: some harm the liver or kidneys, some cause cancer; how much is left depends on the mould

→ Heat kills the mould but not its toxins; nobody has measured how much is left after scraping

### 53 Charred barbecue meat

↑ They are considered carcinogens, but one meal does no harm

→ Most people in Latvia barbecue regularly, but the black bits are a small part of the meal

### 54 Several supplements at once

↑ The Netherlands received 238 reports of nerve damage linked to vitamin B6

→ In the Baltics over 40 % take 3–5 supplements; what decides is each nutrient's total, not the number of products

### 55 Nitrosamines

↑ To be safe, experts assumed every cancer-causing nitrosamine is as potent as the most potent one. So the real hazard is probably lower

→ Every age group takes in enough nitrosamines to raise a health concern. Meat products are the main source

### 56 Ergot alkaloids

↑ The concern is long-term intake from food, especially in toddlers

→ Cleaning removes the black grains, but alkaloids can remain; intake is highest where a lot of rye bread is eaten

### 57 Pyrrolizidine alkaloids

↑ High doses cause severe liver damage; no safe dose can be set for the cancer risk

→ One cup means nothing; the concern is for people who drink a lot of one tea for years

### 58 Tropane alkaloids

↑ Hallucinations and confusion set in within 5–30 minutes, and a doctor may be needed

→ Buckwheat is eaten often, but thorn-apple seeds rarely get into the harvest, and since 2022 there is a European Union limit

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### 59 Lead from petrol

↑ Swedish children averaged 60 µg/L of lead in their blood in 1978: five times the level that already costs one intelligence quotient (IQ) point

→ Lead came from air and food at once, so everyone's total dose was high

### 60 Methylmercury in Minamata

↑ By August 1956 there were 30 cases, 11 of them fatal. In the worst-hit areas 13 of 188 babies born in 1955–1958 were severely ill

→ For 36 years factory waste carried methylmercury into the bay, fish built it up, and families ate them daily

### 61 Cadmium in Toyama rice

↑ Worst hit were women over 35 who had given birth: bones broke until the person could no longer move

→ Cadmium builds up in the kidneys, and rice and river water brought it in every day for decades

### 62 Melamine in infant formula

↑ By the end of November 2008, 294,000 babies were affected, over 50,000 were in hospital and six died

→ Formula was the main food, so a baby on the most contaminated brand took in 40–120 times the tolerable intake

### 63 Ergotism epidemic

↑ Four Ural districts registered 10,104 cases; 1,633 went to hospital and 84 of them died

→ The poorest had no old grain, so they ate the fresh harvest, when ergot is most toxic

### 64 Polyols (sugar alcohols)

↑ The diarrhoea starts soon after eating and passes on its own

→ Only people who eat sugar-free products take them in, but those who eat a lot exceed the daily limit

### 65 Honey for babies

↑ If a baby falls ill, it usually needs a long hospital stay, often with breathing support

→ Spores are found in about 4 % of honey samples worldwide. Infant botulism is very rare; none in Latvia in 2015–2022

### 66 Home-canned food

↑ The toxin paralyses muscles; it needs hospital care and antitoxin, and recovery takes weeks or months

→ The toxin forms if spores survive in a jar without acid. About 80 cases a year in Europe, none in Latvia in 2015–2022

### 67 Hepatitis E virus

↑ Most infected people notice nothing; only people with liver disease or weak immunity fall seriously ill

→ Relatively common in pork: in Germany its RNA was found in 1 in 5 liver sausages. Thorough cooking kills it

### 68 Anisakis: fish parasites

↑ A live larva causes sharp pain and usually needs a doctor; half of allergy cases need emergency care

→ Larvae are in many sea fish, but live ones rarely reach people: fish for eating raw is frozen, home-salted fish not always

### 69 Chemicals from packaging and tableware

↑ Even for bamboo cups, some of which released melamine and formaldehyde over the limits, the European Commission said: no acute risk

→ Food touches packaging, dishes and utensils every day, so a little always passes into it

### 70 6PPD-quinone

↑ Deadly to salmon, harms the liver and nerves of lab animals; whether it harms people is unknown

→ It is found in human urine and blood, but not in leafy vegetables from Europe and Israel; nobody knows how much comes from food

### 71 Cinnamon and coumarin

↑ Usually nothing happens. Liver damage was seen only in sensitive people taking coumarin as a medicine, and it passed

→ Cassia cinnamon has a lot of coumarin, Ceylon cinnamon little. A 15 kg child reaches the daily limit with 0.5 g of cassia a day

### 72 Well water

↑ An adult usually gets diarrhoea for a few days. Formula made with nitrate-rich water can turn a baby 'blue'

→ Only part of the population drinks well water, mostly in the countryside; there, bacteria were found in five of six wells

### 73 Microwaves

↑ In food, microwaves turn into heat: nothing becomes radioactive, and vitamins are lost no more than in a pot

→ There is one in many kitchens, workplaces and cafés, and many homes use it every day

### 74 Swallowed chewing gum

↑ The gum base really is not digested, but it does not stick. The gut moves it out like other things it cannot digest

→ Gum is swallowed now and then, especially by children. It usually passes within a week

### 75 Swallowed watermelon seeds

↑ A seed needs soil with nutrients. In acidic stomach juice it does not sprout and passes through intact

→ In summer many families eat watermelon, and seeds get swallowed now and then. They pass in about a day

### 76 Swimming straight after eating

↑ In studies, eating did not affect swimming. A meal can bring minor discomfort, not dangerous cramps

→ Now and then in summer people swim soon after eating, but it does not raise the drowning risk. Alcohol is more to blame